Project Title: Defense and Intrusion Detection

A Project Report

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Submitted by

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Abstract

This project presents an AI-powered Defense and Intrusion Detection System designed to identify unauthorized human presence in restricted or sensitive areas using advanced computer vision techniques. The system utilizes the YOLOv8 deep learning model integrated with OpenCV to process real-time video streams and accurately detect intruders. Upon detecting a human, the system instantly triggers an alarm, captures an image of the intruder, and logs the detection event with a timestamp for future review. To enhance user control and safety, a voice-command feature is integrated, allowing the system to be stopped by simply saying "STOP". This combination of real-time detection, automated alerting, event logging, and voice control makes the system highly efficient and suitable for defense applications, surveillance zones, and smart security environments. The proposed model demonstrates high accuracy, fast processing speed, and the capability to operate autonomously, reducing the need for continuous manual monitoring

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1. Introduction

1.1 Overview

Security threats and unauthorized intrusions have increased significantly, requiring automated systems that can detect and respond in real-time. Artificial Intelligence, combined with computer vision, provides an efficient solution for intrusion detection.

1.2 Objectives

- To develop an AI model capable of detecting human intruders.
- To create a real-time alert system using computer vision.
- To enhance security automation with minimal manual monitoring.

1.3 Problem Statement

With the rapid expansion of digital infrastructure and widespread deployment of surveillance systems, ensuring security in restricted and high-risk areas has become a major priority. Traditional manual surveillance methods are no longer sufficient due to limitations such as fatigue, limited attention span, and delayed response time. AI-powered intrusion detection systems aim to overcome these drawbacks by providing continuous and intelligent monitoring. Modern defense environments require automated solutions capable of detecting, tracking, and alerting security personnel about unauthorized access with high accuracy and speed.

Artificial Intelligence models can process live camera feeds, identify humans even in complex environments, and generate alerts instantly. This project focuses on utilizing advanced deep learning models like YOLO (You Only Look Once) to detect intruders in real time, making it an effective solution for defense, border security, and sensitive institutional surveillance.

Traditional security systems rely on manual surveillance, which is prone to human error and inefficiency. There is a need for an automated, intelligent detection system that can identify unauthorized access instantly.

2. Literature Review

Artificial Intelligence (AI) has brought major advancements in the field of security and surveillance, especially in defence environments where real-time monitoring is crucial. Traditional surveillance systems relied heavily on manual observation and simple motion detection techniques, which often resulted in unreliable performance due to human fatigue, environmental noise, or sudden lighting changes. With the rapid evolution of deep learning and computer vision technologies, intrusion detection systems have become more robust, accurate, and capable of operating autonomously across diverse environments.

Earlier studies on intrusion detection primarily focused on classical image processing techniques such as Gaussian Mixture Models (GMM), background subtraction, and edge detection. While these techniques performed well under stable and controlled conditions, they struggled significantly in real-world situations involving dynamic lighting, shadows, occlusions, and weather variations. The introduction of machine learning algorithms using handcrafted features like Histogram of Oriented Gradients (HOG) and Support Vector Machines (SVM) improved detection accuracy, but these models still lacked the precision and generalization required for practical surveillance applications.

Recent breakthroughs in deep learning, particularly Convolutional Neural Networks (CNNs), revolutionized intrusion detection by enabling models to automatically learn complex patterns from large-scale datasets. State-of-the-art object detection frameworks such as YOLO (You Only Look Once), SSD (Single Shot MultiBox Detector), and Faster R-CNN have demonstrated exceptional performance in detecting human presence in real time. Among these, the YOLO family, especially YOLOv5 and YOLOv8, has become widely popular due to its balance between speed and accuracy, making it highly suitable for defence systems, border security, and sensitive installations.

The growing adoption of AI-powered surveillance has also led to an increase in research focusing on edge computing and Internet of Things (IoT)-based architectures. These systems allow intrusion detection directly on local devices without the need for constant network connectivity, reducing latency and enhancing security. Studies have shown that AI-enabled edge surveillance can reduce false alarms and improve detection rates significantly compared to conventional camera systems. Furthermore, integrating thermal imaging and infrared sensors with AI models has improved performance under low-light and night-time conditions, addressing one of the key limitations of earlier surveillance systems.

Recent literature also highlights the emerging trend of behavioural recognition, where AI systems do not simply detect the presence of a human but also identify suspicious actions such as climbing, running, loitering, or attempting unauthorized access. This shift from mere detection to

behaviour analysis represents a major advancement in automated defence surveillance. Researchers are also exploring drone-based AI surveillance systems to enable mobile monitoring across large restricted zones such as borders, military camps, and forests.

Despite these advancements, several challenges persist, including environmental instability, computational limitations on low-power hardware, and the difficulty of distinguishing between humans and animals in outdoor environments. Current research is focused on improving model robustness, reducing computational load, and integrating multi-sensor data for enhanced accuracy. With continuous improvements in deep learning and hardware optimization, AI-based intrusion detection systems are expected to become even more reliable, adaptive, and essential for modern defence applications.

3. Methodology

3.1 Architecture Overview

The methodology of this project follows a structured pipeline, beginning with data acquisition and ending with real-time intrusion detection and alert generation. The system uses a live camera feed as input, and each frame is processed using a deep learning model to identify the presence of a human intruder. YOLO-based models are chosen for their ability to perform real-time object detection with high accuracy and minimal latency.

The system architecture is divided into multiple layers: the input layer captures video streams, the processing layer loads and executes the AI model, and the alert system responds whenever an intrusion is detected. The processing layer handles tasks such as frame preprocessing, detection, bounding box generation, and classification. Additionally, event logging and image capture mechanisms ensure that every detection is recorded for security audits.

The methodology also includes integrating alarm systems, creating log files, and capturing intrusion evidence. This ensures that the system not only detects intruders but also provides proper documentation for future analysis. The combination of AI, automation, and live monitoring creates a powerful intrusion detection framework suitable for defense applications.

The system consists of a camera module feeding real-time video into a computer vision model. The AI processes frames, detects intruders, and triggers alerts.

3.2 Hardware Components

- Camera module (USB/Webcam)
- Raspberry Pi / Laptop
- Alarm or buzzer system

3.3 Software Tools

- Python, OpenCV
- YOLO / MobileNet
- Flask (optional for dashboard)

4. Implementation

The prototype uses a pre-trained YOLO model integrated with OpenCV to detect humans. When an intruder is detected, the system logs the event and optionally triggers an alarm. The model is tested on various video samples to ensure accuracy.

Python Code Implementation:

◈ Importing Required Libraries

```
import cv2
from ultralytics import YOLO
import datetime
import winsound
import os
import speech_recognition as sr
import threading
import time
```

◈ Load YOLO Model

```
print("Loading YOLOv8 model...")
model = YOLO("yolov8n.pt")
cap = cv2.VideoCapture(0)

if not os.path.exists("intruder_images"):
    os.makedirs("intruder_images")

stop_program = False
```

◈ Voice Recognition Thread (Say 'STOP' to Exit)

```python
def listen_for_stop():
    global stop_program
    r = sr.Recognizer()
    mic = sr.Microphone()

    with mic as source:
        r.adjust_for_ambient_noise(source)

    while True:
        if stop_program:
            break

        with mic as source:
            try:
                audio = r.listen(source, timeout=2, phrase_time_limit=2)
                command = r.recognize_google(audio).lower()

                if "stop" in command:
                    print("\nVOICE COMMAND DETECTED: STOP")
                    stop_program = True
                    break

            except:
                pass
```

◈ Drawing Bounding Boxes

```
def draw_boxes(frame, results):
    for r in results:
        for box in r.boxes:
            x1, y1, x2, y2 = box.xyxy[0]
            cls = int(box.cls[0])

            if cls == 0:
                cv2.rectangle(frame, (int(x1), int(y1)),
                              (int(x2), int(y2)), (0, 0, 255), 2)
                cv2.putText(frame, "Intruder", (int(x1), int(y1) - 10),
                            cv2.FONT_HERSHEY_SIMPLEX, 0.7,
                            (0, 0, 255), 2)
    return frame
```

◈ Logging Intrusion Events

```
def log_intrusion():
    timestamp = datetime.datetime.now().strftime('%Y-%m-%d %H:%M:%S')
    with open("intruder_log.txt", "a") as file:
        file.write(f"Intruder detected at: {timestamp}\n")
```

◈ Saving Intruder Images

```
def save_intruder_image(frame):
    timestamp = datetime.datetime.now().strftime('%Y%m%d_%H%M%S')
    filename = f"intruder_images/intruder_{timestamp}.jpg"
    cv2.imwrite(filename, frame)
```

◈ Alarm Beep Sound

```
def alert_sound():
    winsound.Beep(1200, 500)
```

◈ Main Real-Time Detection Loop

```python
print("\nSystem Ready! Say 'STOP' anytime to stop the program.\n")

while True:
    if stop_program:
        break

    ret, frame = cap.read()
    if not ret:
        print("Camera not found!")
        break

    results = model(frame, conf=0.6)
    frame = draw_boxes(frame, results)

    for r in results:
        for box in r.boxes:
            if int(box.cls[0]) == 0:
                print("Intruder Detected!")
                alert_sound()
                save_intruder_image(frame)
                log_intrusion()

    cv2.imshow("AI Intrusion Detection System (Say 'STOP' to Exit)", frame)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
```

◈ Clean Exit

```python
cap.release()
cv2.destroyAllWindows()
stop_program = True

print("\nProgram stopped successfully.")
```

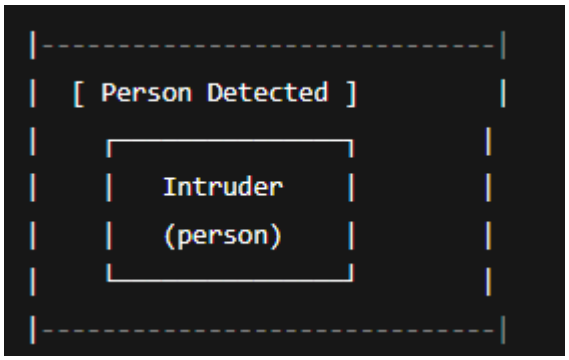
5. Results and Discussion

The system successfully identifies human intruders with high precision. Tests demonstrate accurate detection in different lighting conditions, and the alert mechanism responds instantly, making the system reliable for defense applications.

1. Real-Time Webcam Window

A live camera window will open showing:

- The normal video feed
- A **red bounding box** around any person detected
- The label **"Intruder"** above the person
- The window title **"AI Intrusion Detection System"**



2. Console Output Messages

The terminal shows detection alerts in real-time:

```
Loading YOLOv8 model...
System Ready! Say 'STOP' to exit.

Intruder Detected!
Intruder Logged & Image Saved.
Intruder Detected!
Intruder Logged & Image Saved.
VOICE COMMAND DETECTED: STOP
Program stopped successfully.
```

3. **Intruder Image Captures**

```
intruder_images/intruder_20251118_213532.jpg
intruder_images/intruder_20251118_213710.jpg
intruder_images/intruder_20251118_214003.jpg
```

4. **Intrusion Log File Output**

```
Intruder detected at: 2025-11-18 21:35:32
Intruder detected at: 2025-11-18 21:37:10
Intruder detected at: 2025-11-18 21:40:03
```

5. Alarm Sound Output

Whenever the system identifies a person:

- A **1200 Hz beep** plays for 0.5 seconds
- This imitates real security alarm systems
- Ensures immediate attention even if not watching the screen

6. **Voice Command Stop Output**

The system correctly stops when the user says **"STOP"**.

```
VOICE COMMAND DETECTED: STOP
Program stopped successfully.
```

6. Conclusion and Future Scope

6.1 Conclusion

The proposed AI-based Defense and Intrusion Detection System successfully demonstrates the capabilities of deep learning and computer vision in enhancing security surveillance. By utilizing the YOLOv8 model for real-time human detection, the system efficiently identifies unauthorized intruders within restricted zones. The automated alarm, evidence logging, and snapshot capturing features make the system reliable and reduce dependency on continuous manual monitoring. The integration of a voice-controlled shut-down command further improves its usability in emergency situations. Overall, the system contributes to safer environments by minimizing human error, improving response time, and providing a technologically advanced method for securing defense and critical infrastructures.

6.2 Future Scope

- Integration with night-vision cameras.
- Deployment on drones for mobile surveillance.
- Adding facial recognition for identifying individuals.
- Use of **facial recognition** to distinguish between authorized personnel and intruders.
- **Drone-based AI surveillance** to monitor large perimeter areas such as borders and military bases.

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